Why The Average Lies

Two processes can share the exact same average and behave in completely opposite ways.

The Four-Hour Lie

A support team reported an average ticket resolution time of four hours, well under the six-hour target, so leadership left them alone. But a customer complained, so the team built a histogram. The shape told a completely different story than the number. The data was not one tidy hump centered on four hours.

Two Humps

It had two humps. A large cluster near one hour, quick password resets and simple questions. A smaller cluster way out around eleven and twelve hours. Average a pile of one-hour tickets with a pile of eleven-hour tickets and you get four. Nobody was actually having a four-hour experience. Customers were either delighted or furious.

Reading Shapes

Look at the center, the spread, and the shape. A single symmetric hump is the stable bell. A long tail to one side, called skew, means most cases are fine but some drag out. Two humps, called bimodal, almost always means two things are mixed together. The eleven-hour tickets all needed a specialist who worked only part time.

Remember this

Same average, wildly different experience, and only the histogram can tell them apart.